Gabe Schoenbach

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Department of Computer Science
University of Chicago

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1 Education

- PhD, Computer Science, University of Chicago — 2022–Present
  Advisor: Aloni Cohen.
- MS, Computer Science, University of Chicago — 2025
  Committee: Aloni Cohen, David Cash, William Hoza.
- BA, Mathematics, University of Chicago — 2020

Research Interests

- Theoretical Computer Science: Average-case complexity theory, computation-to-optimization gaps, Markov chains for sampling graph partitions.
- Cryptography: Public-key encryption, searchable encryption, watermarks for language models.

Awards and Fellowships

- Summer Fellow, Voting Rights Data Institute at Tufts University — 2019
- UChicago Dean's List, Odyssey Scholar, and University Scholar Award — 2016–2020
- National Merit Scholar — 2016

Research Experience

- *Graduate Research Assistant*, University of Chicago — 2022–Present
  Advised by Aloni Cohen
- *Research Fellow*, Tufts University — 2020-2022
  MGGG Redistricting Lab, hosted by Moon Duchin

Publications

[1] Alexander Hoover, Ruth Ng, Gabe Schoenbach, Keer Guan, Claire-Leia Ng, and Richard Ong. Lower Bounds and Flexible Constructions for Encrypted Range Search. *In submission*, 2026.

[2] Hugo A. Akitaya, Sarah Cannon, Gregory Herschlag, Gabe Schoenbach, Kristopher Tapp, and Jamie Tucker-Foltz. The Balanced Up-Down Walk. *In submission*, 2026.

[3] David Gamarnik, Miklós Z. Rácz, and Gabe Schoenbach. Optimal Hardness of Online Algorithms for Large Common Induced Subgraphs. *Conference on Learning Theory*, 2026.

[4] Aloni Cohen, Alexander Hoover, and Gabe Schoenbach Watermarking Language Models for Many Adaptive Users. *IEEE Symposium on Security & Privacy*, 2025.

[5] Moon Duchin and Gabe Schoenbach. Redistricting for proportionality. In *The Forum*, volume 20, pages 371–393. De Gruyter, 2023.

[6] Daryl DeFord, Natasha Dhamankar, Moon Duchin, Varun Gupta, Mackenzie McPike, Gabe Schoenbach, and Ki Wan Sim. Implementing partisan symmetry: A response to a response. *Political Analysis*, 31(3):332–334, 2023.

[7] Daryl DeFord, Natasha Dhamankar, Moon Duchin, Varun Gupta, Mackenzie McPike, Gabe Schoenbach, and Ki Wan Sim. Implementing partisan symmetry: Problems and paradoxes. *Political Analysis*, 31(3):305–324, 2023.

[8] Karin C Knudson, Gabe Schoenbach, and Amariah Becker. Pyei: a python package for ecological inference. *Journal of Open Source Software*, 6(64):3397, 2021.

Talks

- **Optimal Hardness of Online Algorithms for Large Common Induced Subgraphs** 2026
  Conference on Learning Theory, NYU Student Probability Seminar, Columbia Theory Student Seminar, UChicago Theory Lunch Seminar
- **Watermarking Language Models for Many Adaptive Users** 2025
  IEEE Symposium on Security & Privacy, Berkeley Security Seminar
- **Range-Searchable Symmetric Encryption with Minimal Communication Complexity** 2025
  UChicago Masters' Presentation
- **Combinatorial Properties of Traceable Codes** 2023
  UChicago Theory Lunch Seminar
- **County-Aware Redistricting** 2023
  UChicago Theory Lunch Seminar
- **A Workflow for Testing EI** 2021
  MGGG Lab Seminar
- **Counting Graph Partitions** 2021
  MGGG Graph Algorithms Seminar
- **EI Considerations** 2020
  MGGG Lab Seminar
- **Exploring Partisan Symmetry** 2020
  UChicago Undergraduate Math Club

Professional Service

- **External Reviewer**
  - La Matematica 2026
  - Usenix Security Symposium 2026
  - SIAM Journal on Computing (SICOMP) 2026
  - Eurocrypt 2026
  - International Conference and Workshops on Algorithms and Computation (WALCOM) 2026
  - Crypto 2025
  - Foundations of Responsible Computing (FORC) 2023-2024
  - Computer and Communications Security (CCS) 2023
- **Conference/Workshop Volunteer**
  - Theory and Practice of Differential Privacy (TPDP) 2023
  - ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT) 2023

- **UChicago CS Theory Lunch Seminar** January 2024–March 2025
- **Teaching Assistant**
  - Theory of Algorithms (CMSC 27200) Spring 2026
  - Introduction to Cryptography (CMSC 28400) Autumn 2024, Winter 2025
  - Discrete Mathematics (CMSC 27100) Winter 2025, Autumn 2025
  - Advanced Algorithms (MPCS 55005) Spring 2025